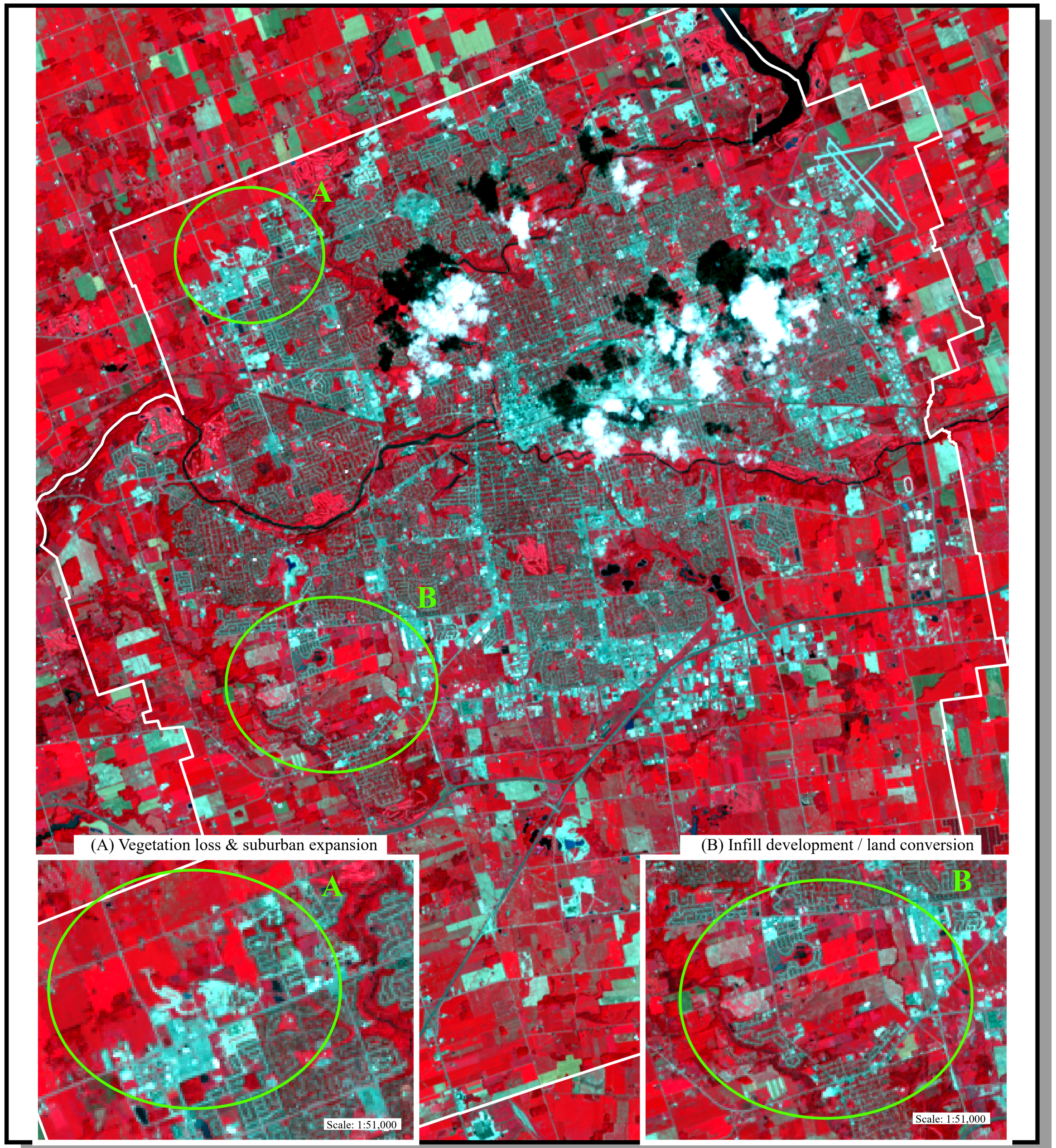
Vegetation and Urban Expansion in London, Ontario (2015–2025): False Colour Analysis

Map Creator: Connor Payne
Data Sources: USGS (EarthExplorer) - (Landsat 8, 2015 & Landsat 9, 2025) & City of London Open Data Portal (2019)
Projection: WGS 1984 UTM Zone 17N
Date: 2026-04-19

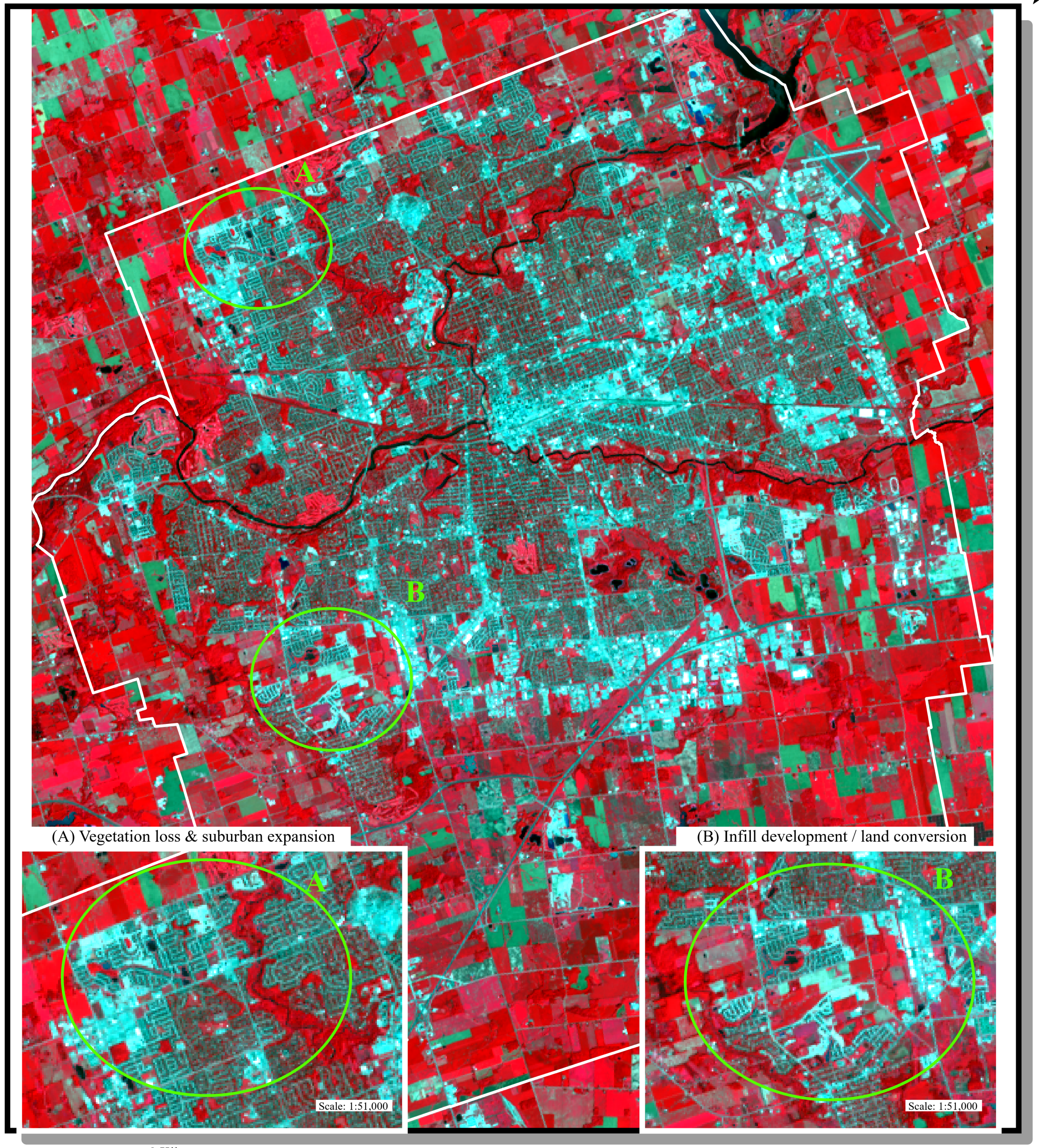
Scale = 1:95,000

N

London, Ontario (2015)



London, Ontario (2025)



Non Built-Up / Natural Areas
Built-up / Urban Areas
Water Bodies

0 2 4 8 Kilometers

Data Sources and Methods

False colour satellite imagery was obtained from the United States Geological Survey (USGS) EarthExplorer platform using Landsat 8 (2015) and Landsat 9 (2025) Operational Land Imager (OLI) data. Images acquired on July 28, 2015 and July 15, 2025 were selected to ensure consistent seasonal conditions and minimize variation in vegetation and illumination. Both datasets correspond to WRS-2 Path 19 and Row 30, covering the study area of London, Ontario.

Processing was conducted in ArcGIS Pro using the Composite Bands tool. A false colour composite was created using Bands 5 (Near Infrared), 4 (Red), and 3 (Green). In this configuration, natural areas appears in bright red tones due to strong reflectance in the near-infrared band, while built-up and impervious surfaces appear in light blue or cyan tones, and water bodies appear dark. The imagery was clipped to the City of London boundary using municipal open data and projected to WGS 1984 / UTM Zone 17N. A percent clip stretch was applied to enhance contrast and improve visualization of land cover differences.

To support interpretation of spatial change, specific areas of interest were identified and highlighted on both the 2015 and 2025 maps using labelled callouts (A and B). Insets were created for these locations to provide a more detailed view of land cover change and allow for clearer comparison between time periods.

Results

The false colour imagery provides a clearer distinction between non built-up and built-up areas compared to the true colour composite, making patterns of land cover change more visually apparent. In the 2015 image, extensive areas of bright red indicate widespread vegetation and agricultural land surrounding the city. By 2025, these red areas are noticeably reduced, particularly along the urban fringe, where they have been replaced by light blue and cyan tones representing built-up and impervious surfaces.

The highlighted areas further illustrate these changes. In Area A, located along the urban periphery, there is a clear transition from predominantly undeveloped land in 2015 to developed suburban areas in 2025. This change is emphasized in the inset, where the reduction in red vegetation and expansion of cyan-toned urban features are clearly visible. In Area B, the imagery shows both outward expansion and localized infill development, with new construction occurring within and adjacent to existing urban areas.

These changes are more easily identified in the false colour composite compared to the true colour imagery due to the enhanced contrast between non built-up and built-up land. The inset maps provide additional detail, making it easier to observe patterns of land conversion that may be less visible at the full map scale. Overall, the results indicate a decrease in undeveloped land and a corresponding increase in urban development across the study area between 2015 and 2025.